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Personal Portfolio & Dynamic CV Architecture Blueprint

Target Platform: GitHub Pages

Tech Stack: Astro 5 + Tailwind CSS v4 + GSAP

Design Paradigm: Apple HIG Minimalist

Executive Summary: This document outlines the complete architectural spec for a high-impact, Apple-inspired Hardware & Embedded Engineering Portfolio. By combining a minimal CV interface with a scroll-driven *"Concept-to-Production"* storytelling Hero animation, the platform presents high-level technical mastery without visual clutter.

1. Concept Evaluation & Positioning Strategy

Most engineering portfolios fall into two extremes: overly static text resumes or generic corporate-style websites. The proposed architecture bridges this gap by functioning as a hybrid **Interactive Portfolio + Live Resume + Technical Documentation Engine**.

- **Instant Value Proof:** Highlighting full product lifecycle capabilities (Schematic → PCB Layout → DFM/Assembly → Firmware → Mechanical Enclosure) in under 10 seconds via scroll animation.
- **Intentional Simplicity:** Clean typography and white space signal confidence and clarity, proving that minimalist design is an aesthetic choice rather than a lack of technical depth.
- **Engineered Precision:** Optimized for zero runtime bloat, lightning-fast static delivery, and complete automation via GitHub Actions.

2. System Architecture & Tech Stack

Layer	Technology	Rationale
Framework	Astro 5.0 (SSG)	Zero JS by default, instant page transitions, native Markdown / Content Collections support.
Styling	Tailwind CSS v4	Utility-first, minimal CSS footprint, native CSS container queries, dark mode support.
Interactivity & Animation	GSAP + ScrollTrigger / HTML5 Canvas	Hardware-accelerated scroll scrubbing for the concept-to-production storytelling sequence.
Content Engine	Astro Content Collections	Strictly typed Markdown/MDX schemas for hardware projects, datasheets, and BOM specifications.
CI/CD & PDF Pipeline	GitHub Actions + Typst / Puppeteer	Auto-compiles latest Markdown/JSON CV data into an optimized resume.pdf on git push.
Hosting	GitHub Pages + Custom Domain	100% free, automated deployment, SSL included, ultra-fast global CDN.

3. Scroll-Driven Storyboarding: "Concept to Production"

The Hero section replaces traditional headshots with an interactive hardware development timeline. As the user scrolls down, the visual asset smoothly morphs across 7 key engineering stages:

Stage Breakdown & Visual Asset Sequence

1. **Stage 1 (Concept & Schematic):** Abstract SVG circuit lines and schematic symbols draw onto a clean canvas. Text: *"Every product starts with an architecture."*
2. **Stage 2 (PCB Layout):** Lines morph into copper traces, power planes, differential pairs, and vias. Text: *"Precision PCB Routing & Signal Integrity."*
3. **Stage 3 (Bare Board):** Gerber layer renders into a high-res 6-layer bare PCB image. Text: *"Production-Ready DFM Design."*
4. **Stage 4 (Component SMT Assembly):** Active components (ESP32, Ethernet PHY, DC/DC Buck, Connectors) drop onto pads with subtle metallic reflections.
5. **Stage 5 (Firmware & Activation):** LED indicators glow green, simulating firmware bootloader flash and peripherals initialization. Text: *"Embedded Firmware & Protocols (C/C++, FreeRTOS, BLE, CAN)."*
6. **Stage 6 (Enclosure & Mechanical Fit):** Industrial CNC/Aluminum enclosure halves slide in from top and bottom, enclosing the board.
7. **Stage 7 (Finished Product & Shrink):** Final product scales down gracefully and docks into the top header navigation bar, seamlessly unlocking the full CV content below.

Technical Implementation Strategy for Animation

To ensure smooth 60fps performance on mobile devices and low-power hardware:

- **Option A (Recommended): Canvas Frame Sequence / WebP Stack:** Pre-render high-quality isometric renders into a 30-frame sequence. GSAP ScrollTrigger scrubs the current frame on HTML5 <canvas>. Lightweight (~1.5MB total WebP sequence), bulletproof performance.
- **Option B (Vector SVG Morph):** Layered SVG vector graphics controlled via CSS opacity and transform. Zero external image assets required, ultra-crisp at any resolution.
- **Mobile Fallback:** Responsive breakpoint switches to a clean swipeable card stack or simplified 3-stage SVG sequence on screens < 768px to conserve mobile GPU resources.

4. Automated PDF Resume Pipeline

To ensure the downloadable resume.pdf is always 100% synchronized with the website content:

```

.github/workflows/build-pdf.yml
-----
name: Build and Deploy Portfolio & Resume
on:
  push:
    branches: [ main ]

jobs:
  build-resume:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - name: Setup Node.js & Astro
        uses: actions/setup-node@v4
        with:
          node-version: 20
      - name: Generate PDF Resume via Typst / Playwright
        run: npx typst compile src/content/resume.typ public/resume.pdf
      - name: Build Astro Static Site
        run: npm run build
      - name: Deploy to GitHub Pages
        uses: peaceiris/actions-gh-pages@v3
        with:
          github_token: ${{ secrets.GITHUB_TOKEN }}
          publish_dir: ./dist

```

5. Project Data Structure & Directory Architecture

```

my-portfolio/
├── .github/workflows/deploy.yml
├── public/
│   ├── favicon.ico
│   ├── resume.pdf (auto-generated)
│   └── assets/projects/
├── src/
│   ├── components/
│   │   ├── ConceptHeroAnimation.astro
│   │   ├── ProjectCard.astro
│   │   ├── Navigation.astro
│   │   └── Footer.astro
│   ├── content/
│   │   ├── config.ts
│   │   └── projects/
│   │       ├── hpm20.md
│   │       ├── esp32-lte-tracker.md
│   │       └── poe-power-supply.md
│   ├── layouts/
│   │   ├── BaseLayout.astro
│   │   └── ProjectLayout.astro
│   └── pages/
│       ├── index.astro (CV / Main Landing)
│       └── projects/
│           └── [slug].astro (Interactive Detail View)
└── astro.config.mjs

```

6. Design Language Specifications (Apple HIG)

Light Theme Palette

- **Background:** #FBFBFD
- **Card Surface:** #FFFFFF
- **Primary Text:** #1D1D1F
- **Secondary Text:** #6E6E73
- **Accent Color:** #0071E3
- **Border:** #D2D2D7

Dark Theme Palette

- **Background:** #000000
- **Card Surface:** #1C1C1E
- **Primary Text:** #F5F5F7
- **Secondary Text:** #A1A1A6
- **Accent Color:** #2997FF
- **Border:** #3A3A3C

7. Roadmap & Next Steps

1. **Phase 1: Repo Setup & Content Schema:** Initialize Astro project, configure Tailwind CSS v4, and create JSON/Markdown schemas for Experience, Skills, and Projects.
2. **Phase 2: Core CV Layout:** Build the Apple-style single-page resume layout (Hero, Work Experience, Technical Skills grid, Education, Featured Projects).
3. **Phase 3: Concept-to-Production Animation:** Implement GSAP ScrollTrigger canvas/SVG sequence in the Hero section.
4. **Phase 4: Dedicated Project Pages:** Design standard template for hardware project documentation (Overview, PCB specs, Block diagrams, Downloads).
5. **Phase 5: GitHub Actions PDF Automation:** Configure pipeline to render Markdown to PDF and auto-deploy to GitHub Pages.